1 Processing of Merged Operations

Purpose

Merged operations are a special form of the serial production. In the relevant planning level (e.g. in HYDRA shop floor scheduling) or on the shop floor terminal, you combine different operations with a short run time each to a logic group, which then has a manageable run time. This group of operations is called "merged operation". The system creates a "substitute" operation representing the merged operation. You then plan and log on only the "substitute" operation that contains all single operations. You use different configurable perspectives to distribute the data recorded.

Integration

To create a merged operation, HYDRA provides two methods:

Creating merged operations on the terminal

The operator creates the merged operation on the terminal. When you log on a merged operation, you enter the single operations one after the other and assign them to the relevant person. When you log off or interrupt this merged operation, you must only enter the person.

Creating merged operations on the MOC

The function *Generate merged operation* combines separate operations and builds a substitute operation that stands in for all the other short operations. You log on the substitute operation on the terminal.

If you have created a merged operation on the MOC, you can schedule this merged operation using the MOC planning functions. This is not possible with merged operations created on the terminal.

Requirements

You cannot create merged operations on the terminal and on the MOC. This is incompatible. You must configure in the HYDRA basic settings which one of the two methods is used. Subject to this configuration, the functions for merged operations are available either on the terminal or on the MOC only.

1.1 Creating merged operations on the terminal

Using the function *Merged operation* on the terminal, you can combine several operations and build one merged operation (MOP) on the terminal.

Order processing using merged operations is useful if:

- several operations with short run times are combined to build one operation (lower posting efforts),
- several operations are produced using one workplace at the same time. The system cannot make timely logon/logoff postings for the separate operations (for example during hardening in the metal industry or smoking in the food industry).

1.1.1 Different posting types of merged operations

The below-mentioned types of postings for merged operations are specified in the Terminal configuration.

"Generation per person"

When you log on a MOP, you log on the combination of person and orders. The person who combines several operations to build a MOP is automatically logged on to all separate operations included in the MOP.
Only one MOP is possible per person. If a person performs another MOP logon at a later time, then these separate operations are automatically assigned to the already existing MOP.
A MOP can only be logged on, logged off or interrupted. You cannot log on additional persons to a person-related MOP.

"Generation per machine"

With this configuration, you can use the function *Log on merged operation* on the terminal to combine several OPs and build one MOP. Per machine, you can create one MOP.
Just as it is the case for single operations, several persons can log on to this MOP. If you interrupt or log off the MOP, all persons logged on are automatically logged off. The posting of quantities and the distribution of times is made in accordance with the specifications described in the sections that follow.

1.1.2 Posting of merged operations on the terminal

For information on how to create merged operations on the terminal, refer to the relevant terminal documentation *(for Windows or DOS terminals)*.

Merged operations are created on the terminal as described in the relevant documentation. The terminal generates a substitute order number for the merged operation. Depending on the posting type, the user's badge number or the machine/workplace number is integrated in this order number:

Merged operation number for the "generation per person" option

Depending on the length of the order number and staff badge number configured in the system, the number of the merged operation is created as follows:

The length of the order no. is at least four characters longer than the staff badge no.:

SAM-XXXXX xxxxx represents the staff badge no.

The length of the order no. is at least two characters longer than the staff badge no.:

S-XXXXX xxxxx represents the staff badge no.

The order number length is **not** at least two characters longer than the staff badge number:

In this case, the standard function "merged operations on the terminal" cannot be used.

Merged operation number for the "generation per machine" option

Depending on the length of the order number and the machine number configured in the system, the number of the merged operation is created as follows:

Length of the order number is at least two characters longer than the length of the machine/workplace number:

S-XXXXX xxxxx represents the machine/workplace number

Length of the order number is at least four characters longer than the length of the machine/workplace number:

SAM-XXXXX xxxxx represents the machine/workplace number

The length of the order number is **not** at least two characters longer than the length of the machine/workplace number:

In this case, the standard function "merged operations on the terminal" cannot be used.

1.1.3 Booking of merged operations created on the terminal

The actual times recorded for the MOP can be distributed according to the following methods:

According to the **standard times**, i.e. in relation to the standard times of the separate OPs

According to the **default quantities**, i.e. in relation to the target quantities of the separate OPs

According to the **individual OP**, i.e. according to the number of the logged on single OPs

The posting method is defined in the Basic settings.

The options "according to standard time" or "according to default quantity" must not be used together with the option *Proportionate RPA posting in personnel postings* in the basic settings.

Please contact MPDV Support to configure the MOP posting method.

Example - Distribution according to standard time

Distribution of the actual times of the MOP according to the standard times of the separate OPs:

3 operations are combined and form one MOP:

OP	Standard time
OP01	8000 sec
OP02	600 sec
OP03	6000 sec

A log record for the MOP with the run time of 12000 sec must now be distributed to the separate OPs. The following formula is used to distribute the times

$$Posted\,times = \frac{Total\,duration\,of\,the\,\log\,record * standard\,time\,of\,individual\,OP}{Total\,of\,the\,standard\,times\,of\,all\,OPs}$$

The following values result for the separate OPs:

OP	Run time booked
OP01	6575 sec
OP02	493 sec
OP03	4932 sec

Sample calculation for OP 01:

Sum of standard times of all separate OPs = 8000 + 600 + 6000 = 14600

Run time booked = 12000 * 8000/14600 = 6575 sec

PLEASE NOTE:

The standard time of an operation is calculated using the **target setup time + target duration**.

Example - Distribution according to standard time

Distribution of the actual times of the MOP according to the target quantities of the separate OPs:

3 operations are combined to form one MOP:

OP	Target quantity
OP01	1000
OP02	500
OP03	2500

A log record for the merged OP with the run time 12000 sec must now be booked to the different OPs. To distribute the times, the following formula applies

$$Posted\,time = \frac{Total\,duration\,of\,\log record * target\,quantity\,of\,individual\,OP}{Total\,of\,target\,quantities\,of\,all\,individual\,OPs}$$

The following values result for the separate OPs:

OP	Run time booked
OP01	3000 sec
OP02	1500 sec
OP03	7500 sec

Sample calculation for OP01:

Sum of target quantities of all separate OPs = 1000 + 500 + 2500 = 4000

Run time booked = 12000 * 1000/4000 = 3000 sec

Example – Distribution according to the number of separate operations

Distribution of actual times of the MOP according to single OPs

3 OPs are combined to form one MOP:

A log record for the MOP with the run time 12000 sec must now be distributed to the different OPs. To distribute the times, the following formula applies

$$Posted\,time = \frac{Total\,duration\,of\,the\log record}{number\,of\,individual\,OPs}$$

The following values result for the separate OPs:

OP	Run time booked
OP01	4000 sec
OP02	4000 sec
OP03	4000 sec

The **quantities and the times recorded** are also **posted for the person** who processes the MOP.

1.1.4 Notes on the configuration and the processing

Configuration and use

- By default, the “merged operation” function is not active and must be enabled for a specific terminal in the terminal configuration.
- Only the ADE terminal provides MOP functions on DOS terminalsCT56x und CT73x . On Windows terminals, the MOP functions can also be used in "MDE" operation mode.
- To produce a MOP, the option *Logon of several OPs* must be set in the workplace/machine configuration, i.e. it is permitted to log on several operations at this workplace.
- When you create merged operations on the terminal, the option *Max. OPs per person* of the HR master data is used. If the option *Max. OPs per person* is set to =1 for the person, the person can only log on one operation as merged operation.
- A person can use a maximum of 20 separate operations to build a merged operation.
- When you log on a MOP on a DOS terminal Typ CT56x und CT73x , a sequencing list is not supported, i.e. you must enter the order numbers via barcode or manually.
- If the number of pieces is not recorded for a merged operation, the actual quantity produced is set to the target number of pieces of the separate OPs when the MOP is logged off.
- If the quantity is recorded for a merged operation, the quantity entered for each operation is posted to the different operations when the merged operation is interrupted or logged off.
- For merged operations that are logged on to workplaces/machines with automatic recording of quantities, the proportionate quantity is posted to each separate operation. The quantities are distributed according to the same key as it is the case for times. See the sections above.
- When the MOP is logged off/interrupted, you must only enter the badge number (MOP person). All operations included in a person's merged operation are logged off automatically.

Supported automatic functions/data collection options

- Terminate OP when reaching target quantity, only option "Y" (as of SP10/2016)
  When the MOP is interrupted, the system checks if the single OP has reached its target quantity. If yes, the relevant OP is finished instead of interrupted. The system does not automatically log off a single OP from a merged operation, if the target quantity of a single OP is reached or exceeded with an upload of a partial quantity or an automatic recording of quantities.
- Terminate OP instead of interrupting it
  When you interrupt the MOP, the system checks if the single OP is finished and not interrupted.
- Interrupt OP instead of terminating it
  When you log off the MOP, the system checks if the single OP is interrupted and not finished.
- Automatic release of succeeding OP

- Configurable posting behavior with change/end/beginning of shift
  The order type specifies if the behavior is configured for the machine or the person. Special feature: With person-related MOPs, the merged operation is interrupted when the person is logged off, regardless of the OP setting.

Not supported or restricted automatic functions/data collection options

- Terminate OP when reaching target quantity: options U, F and K
- Option *Proportionate posting of machine time with parallel OPs* in the machine configuration because this option performs the proportionate posting that is configured for the MOP.
- Target quantity reached output (is only performed for the total quantity of the MOP)
- Collection of serial number
- Data collection with batch management requirement
- Posting of resources
- With the milestone processing, you must be aware that a single OP of a MOP cannot be unmerged from the MOP when a posting is made for the preceding or succeeding OP.
  Example:
  When you log off an OP, the preceding OP cannot be logged off automatically if the preceding OP is included in a merged OP (person or machine).
  The MOP must be logged off manually.
- Other restrictions exist with the CAQ integration and the data collection for MOPs, for example the automatic logon and logoff of inspection OPs.

Waiting period processing

If you must postdate a MOP logon because of the waiting period processing, the first single operation of this merged operation is postdated. Further OPs that might be added to the MOP at a later time need not be postdated.
If a clock-out is posted in the PZE for the person that processed the MOP, then all included operations are automatically interrupted.

If the waiting period processing function is configured accordingly, it is also possible that the person who processes the MOP is automatically logged on again as soon as they clock-in in HYDRA-PZE.

1.2 Creating merged operations on the MOC

You can use this function to combine separate operations to build merged operations on the MOC.

The functions to create and cancel merged operations are available in the *Order overview* and in the *Order sequencing* dialog (to call the order sequencing, the BDE-FST must be licensed).

If a merged operation is created on the MOC, then the “members” of a merged operation are no longer displayed. Also in the sequencing list on the terminals, the member operations are no longer displayed.

The number of members in a merged operation is not limited. You can only add prepared orders/OPs to merged operations that are not already contained in other merged operations.

You **cannot** integrate merged operations themselves into other merged operations.

All postings of a merged operation on the terminal are made for all "members" of a merged operation. You can log on, interrupt and log off merged operations on the terminal exactly like normal operations.

1.2.1 Creating merged operations on the MOC

The method to create merged operations is described here.

1.2.2 Booking of merged operations (created on the MOC)

To distribute the quantities and times recorded for a merged operation among the included operations, the system supports the "homogeneous" and the "inhomogeneous" method for the merged operations that were not created on the MOC.

Homogeneous merged operations

With a homogeneous merged operation, quantities and times are distributed using the overrun principle: all separate operations are “filled up” one after the other according to the specified target number of pieces. If the production surpasses the quantity planned for the total merged operation, then this excess and the respective times are added to the single operation having the largest order number.

Simplified example:

OP	Target quantity	Target run time
OP01	200	4.0 hours
OP02	450	9.0 hours
OP03	250	4.5 hours

A log record of the merged operation containing the real quantity 500 and real duration 5.0 hours must now be redistributed among the separate operations. The number of pieces is posted to the separate OPs one after the other (up to the specified target quantity). Depending on the posted number of pieces, the following formula is used to calculate the times:

$$Posted\,time = \frac{Total\log record\,time * posted\,quantity}{Total\,quantity\,of\,the\log record}$$

The following values result for the separate OPs:

OP	Posted quantity	Run time booked
Single OP01	200	2.0 hours
Single OP02	300	3.0 hours
Single OP03	0	0.0 hours

Non-homogeneous merged operations

With an **inhomogeneous merged operation,** the quantities and times are posted proportionately to the separate operations included. If excess production takes place, then the excess is distributed among the different operations.

Simplified example:

OP	Target quantity	Standard time
Single OP01	400	8.0 hours
Single OP02	20	0.5 hours
Single OP03	30	1.0 hours

A log record for the merged operation including an actual quantity 200 and a run time 4 hours must now be redistributed to the separate operations. The following formulas apply:

Distribution of times

$$Posted\ time = \left(\frac{Standard\ time\ of\ individual\ OP\ *}{Sum\ total\ of\ all\ individual\ OPs}\right) * Total\ duration\ of\ log\ record$$

Distribution of quantities

$$Posted\ quantity = \left(\frac{Target\ quantity\ of\ individual\ OP}{Sum\ total\ of\ all\ individual\ OPs}\right) * Actual\ quantity\ of\ log\ record$$

The following values result for the separate OPs:

OP	**Posted quantity**	**Run time booked**
OP01	178	3.4 hours
OP02	9	0.2 hours
OP03	13	0.4 hours

Sample calculation (for single OP01)

Run time booked = 8/9.5 * 4 = 3.4

Quantity booked = 400/450 * 200 = 178 pieces.

The standard time of an operation is calculated using the **target setup time + processing time**. All data required for this calculation is contained in the order backlog of the operation.

If uneven values like 2.666 result when the piece number is distributed, the calculated quantity is cut off and 2 pieces are booked for the operation. **Exception:** With the last operation, the difference between quantity posted and quantity already booked is used.

The type used for the merged operation in the entire system is specified once in the HYDRA basic settings using the BDE option *Process merged operations*.

The upload of the **times recorded** to the PPS system is performed according to the definition of the merged operation type – homogeneous or inhomogeneous.

If a member of a merged operation is finished e.g. via ERP interface, then this operation is skipped during the calculations described above.

1.2.3 Further notes

Validation checks

Validation checks, which are performed during the booking process (e.g. target quantity validation for upload of a part quantity), are only performed for the merged operation and not for the separate operations assigned to the merged operation.

Changing OP data

Changing the MOP

If you make changes to the merged operation, the single OPs included are NOT changed.

Changing a member of a merged operation

If you make changes to one of the operations included in a merged operation, the master operation ist not changed.

Whether you can make changes or not depends on the relevant status of the operation (same process as with a normal operation).